

Rosen — §9.4: Closures of Relations

Selected Exercises

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Exercise 1

Let R be the relation on the set $\{0, 1, 2, 3\}$ containing the ordered pairs $(0, 1)$, $(1, 1)$, $(1, 2)$, $(2, 0)$, $(2, 2)$, and $(3, 0)$. Find the

- (a) reflexive closure of R .
- (b) symmetric closure of R .

Exercise 2

Let R be the relation $\{(a, b) \mid a \neq b\}$ on the set of integers. What is the reflexive closure of R ?

Exercise 3

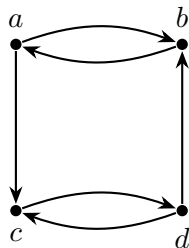
Let R be the relation $\{(a, b) \mid a \text{ divides } b\}$ on the set of integers. What is the symmetric closure of R ?

Exercise 4

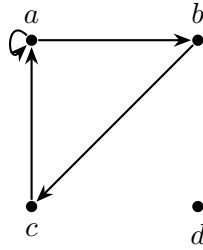
How can the directed graph representing the reflexive closure of a relation on a finite set be constructed from the directed graph of the relation?

Exercise 5

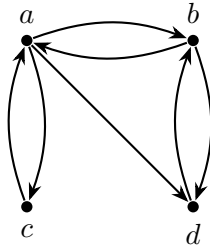
In Exercises 5–7 draw the directed graph of the reflexive closure of the relation with the directed graph shown.



Exercise 6



Exercise 7



Exercise 8

How can the directed graph representing the symmetric closure of a relation on a finite set be constructed from the directed graph for this relation?

Exercise 9

Find the directed graphs of the symmetric closures of the relations with directed graphs shown in Exercises 5–7.

Exercise 10

Recall (Example 2): $R = \{(a, b) \mid a > b\}$ on the set of positive integers.

Find the smallest relation containing the relation in Example 2 that is both reflexive and symmetric.

Exercise 11

Find the directed graph of the smallest relation that is both reflexive and symmetric that contains each of the relations with directed graphs shown in Exercises 5–7.

Exercise 12

Suppose that the relation R on the finite set A is represented by the matrix M_R . Show that the matrix that represents the reflexive closure of R is $M_R \vee I_n$.